EXAMGEN UNIVERSITY

ANNUAL UNIVERSITY EXAMINATION 2026

Course Code: SCI-301 | Examination Year: 2026 | Maximum Marks: **100** | Total Questions: **5** | Time Allowed: **3 Hours**

GENERAL INSTRUCTIONS:

1. All questions are compulsory unless specified otherwise.
2. Candidates must show all working, intermediate steps, and derivations to earn full marks.
3. Draw diagrams wherever necessary; label all parts clearly.
4. Figures, graphs and diagrams are provided with the respective question.
5. Use of scientific calculator is permitted unless otherwise notified.

Q.No	1	2	3	4	5	Total
Marks	20	15	25	20	20	100

Question 1 **[20 Marks]**

Find the derivative of f(x) = sin(x) · e^-0.2x and determine all critical points in the interval [0, 2π]. Sketch the function and clearly label the critical points.

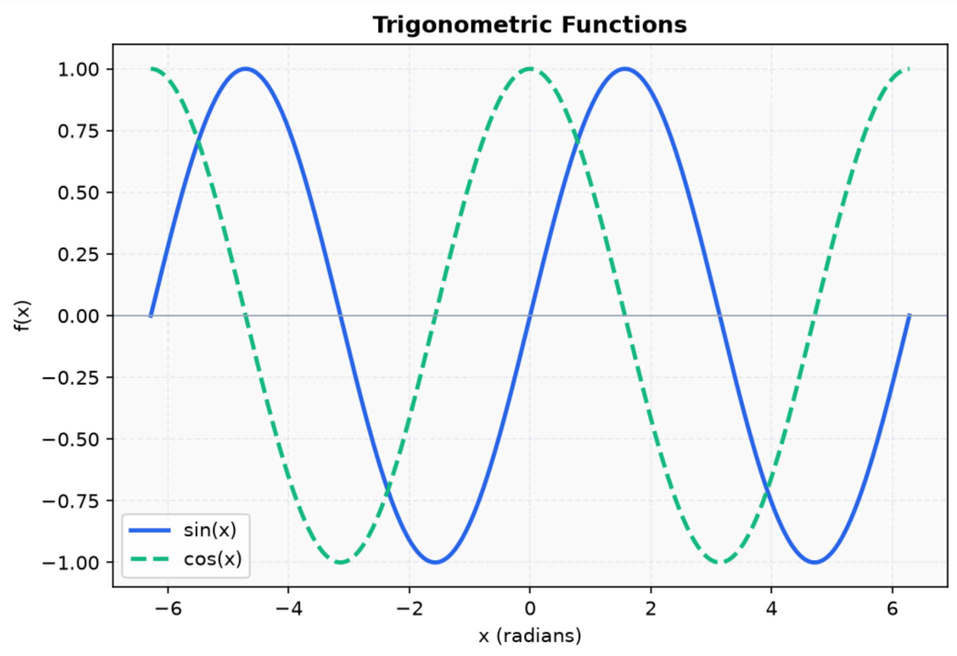

Figure 1

Question 2

[15 Marks]

With reference to the diagram, label the organelles of an animal cell and describe the function of the mitochondria and nucleus. Explain how the cell membrane regulates the entry and exit of substances.

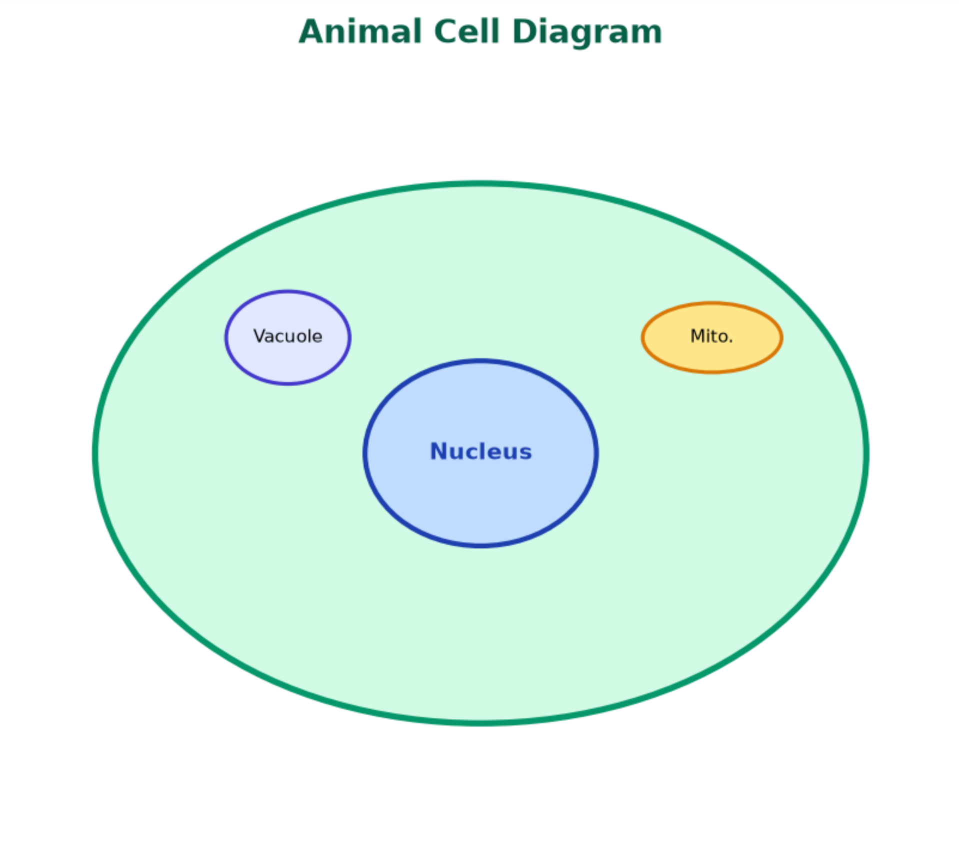

Figure 2

Question 3

[25 Marks]

A block of mass M = 5\,kg is placed on a frictionless horizontal surface. A force F = 20\,N is applied at an angle $\theta = 30°$ to the horizontal. Using Newton's second law F = ma, find the acceleration of the block and the normal reaction force N.

Question 4

[20 Marks]

Evaluate the definite integral ∫[0 to 1] (x^2 / 1+x^3)\,dx using an appropriate substitution. Show all intermediate steps.

Question 5

[20 Marks]

Describe the process of meiosis and distinguish it from mitosis. Include a diagram showing the stages of meiosis I and meiosis II.